UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH
Escola Superior d'Enginyeries Industrial
Aeroespacial i Audiovisual de Terrassa

Bachelor Final Thesis

Document: Project Charter

Bachelor's Degree in Aerospace Vehicle Engineering

Study and Implementation of a Three-Dimensional Numerical Simulator Based on the Finite Element Method in Parallel Computing Environments

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Contents

Contents iii

1 Object 1

2 Scope 3

3 Requirements 5

4 Justification 7

5 Schedule 9

Bibliography 11

Project Charter

1 Object

The object of this project is the study and implementation of a numerical simulator that solves boundary value problems posed on three-dimensional domains and discretised with the finite element method, and that is able to run in parallel on distributed-memory architectures.

Concretely, the simulator addresses the linear advection–diffusion–reaction equation, both in its steady (elliptic) form

$$-\nabla \cdot (a\,\nabla u) + b \cdot \nabla u + c\,u = f \qquad \text{in } \Omega \subset \mathbb{R}^3, \tag{1.1}$$

and in its time-dependent (parabolic) form

$$\frac{\partial u}{\partial t} - \nabla \cdot (a\,\nabla u) + b \cdot \nabla u + c\,u = f \qquad \text{in } \Omega \times (0, T], \tag{1.2}$$

where the diffusion coefficient a, the advection field b, the reaction coefficient c and the source term f may all depend on the point (x, y, z), and equation (1.1) is closed with Dirichlet, Neumann or Robin boundary conditions on the different parts of the boundary $\partial\Omega$.

The implementation is built on top of PETSc, the *Portable, Extensible Toolkit for Scientific Computation* [1], which provides the distributed data structures and parallel solvers on which the simulator relies. Unstructured tetrahedral meshes of the three-dimensional domains are generated externally with Gmsh [2], and the definition of each problem —its coefficients, source terms and boundary conditions— is supplied at run time through an embedded Lua scripting layer, so that a new problem can be solved without recompiling the code.

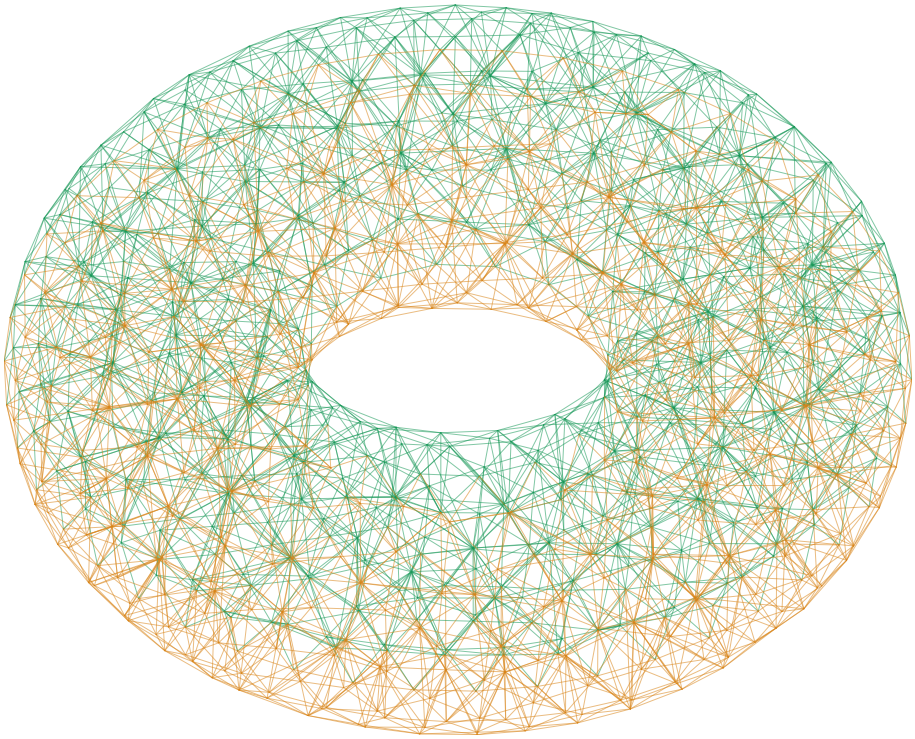


Figure 1.1: Unstructured tetrahedral mesh of a solid torus.

This work also builds upon, and extends to three dimensions and to parallel execution, an earlier two-dimensional finite element solver developed within the same line of work by Gómez [3].

Project Charter 2

Scope

The scope of the project comprises the following tasks:

- The mathematical formulation of the steady and time-dependent advection–diffusion–reaction problems in three dimensions, including their weak (variational) form and the treatment of Dirichlet, Neumann and Robin boundary conditions.
- The finite element discretisation of these problems on unstructured tetrahedral meshes, using Lagrange elements, and the assembly of the resulting discrete residual and Jacobian.
- The import and parallel distribution of three-dimensional meshes generated with Gmsh, together with the identification of the different parts of the boundary to which different boundary conditions are applied.
- The implementation of the simulator in the C language on top of PETSc, using its `DMPlex` mesh abstraction, its `PetscFE` finite element support, and its `KSP`, `SNES` and `TS` solver frameworks.
- The design of a specification layer, based on the Lua language, that lets the user describe a problem —coefficients, source terms and the type and data of each boundary condition— without modifying or recompiling the C code.
- The validation of the simulator against problems with a known analytical solution and against reference solutions computed with the FreeFEM package [4] on the same geometries, together with the assessment of its parallel behaviour by comparing sequential and parallel executions.

Out of Scope

The following aspects are explicitly considered outside the scope of this project:

- The generation of three-dimensional tetrahedral computational grids themselves is delegated externally to Gmsh.
- The study and implementation of nonlinear partial differential equations beyond the linear model equations.
- Execution on graphics processing units (GPUs) or other hardware acceleration platforms.
- Automatic or dynamic refinement techniques during the simulation process.
- Physical laboratory experiments or experimental testing.

Project Charter 3

Requirements

The simulator is subject to the following basic requirements:

- It must solve the steady and time-dependent advection–diffusion–reaction equations on three-dimensional unstructured tetrahedral meshes.
- It must support Dirichlet, Neumann and Robin boundary conditions, and must allow different conditions to be imposed on different parts of the boundary of the same domain.
- The type of boundary condition applied to each part of the boundary, as well as all the coefficients and source terms of the problem, must be configurable at run time, without recompiling the code.
- It must run correctly in parallel on distributed-memory machines using the Message Passing Interface (MPI), and produce results that are independent of the number of processes used.
- It must rely exclusively on free and open-source software (PETSc, an MPI implementation, Gmsh and Lua), so that it is portable and reproducible.
- It must write its results in a format that can be read by standard scientific visualisation tools, so that the computed solution can be inspected and post-processed with, for example, ParaView [5].

Project Charter 4

Justification

Three of the reasons that motivate this project deserve to be highlighted.

First, most problems of engineering interest are genuinely three-dimensional. Two-dimensional models are often convenient simplifications, but they cannot capture geometric features such as the cooling channel of a heat sink or the true shape of a mechanical part. Extending a finite element solver to three dimensions, and doing so in a way that scales to realistic problem sizes, is therefore a natural and valuable step.

Second, the size of three-dimensional problems makes parallel computing essential. A modern open-source stack —with PETSc [1] at its core— makes it possible to build scalable parallel solvers without reimplementing the underlying linear algebra, mesh management or Krylov methods [6] from scratch, and this project takes advantage of exactly that infrastructure.

Third, separating the description of a problem from the code that solves it has a clear practical value. By expressing the coefficients, source terms and boundary conditions in a small Lua specification, the same compiled simulator can be applied to many different problems, which makes it a far more useful tool, both in an engineering and in an educational setting.

Project Charter

5 Schedule

The project is structured into sequential and parallel tasks designed to fulfill the requirements and scope of the 3D finite element numerical simulator. Table 5.1 details the task identification, dependencies, and descriptions.

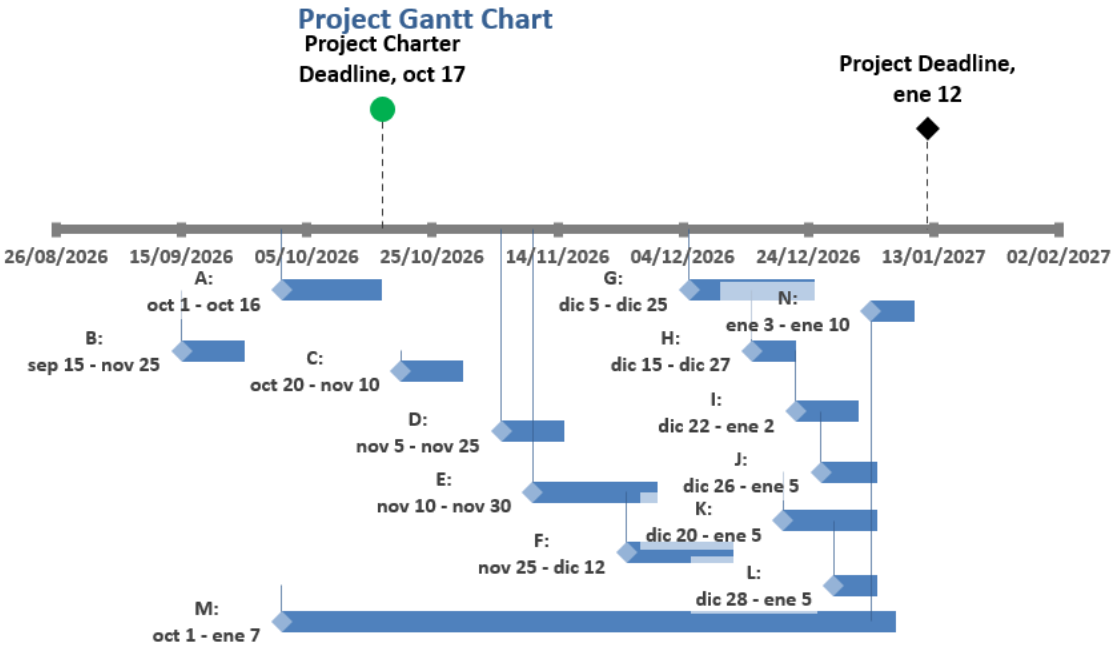


Figure 5.1: Project Gantt Chart (Source: Created by author)

Table 5.1: Project Schedule

Task Code	Identification	Dependencies	Description
A	Project Charter	—	Definition of the project scope, requirements, object, and schedule.
B	Mathematical Foundations Study	—	Study of the 3D boundary value problem, weak formulation, and finite element discretisation on tetrahedra.
C	PETSc & Infrastructure Setup	B	Initial implementation of distributed data structures (`Vec`, `Mat`, `KSP`) and setup of PETSc environment.
D	3D Domain & Gmsh Integration	C	Implementation of the mesh import pipeline using Gmsh (`DMPlex`), parallel mesh distribution (`METIS/PT-Scotch`), and boundary tagging.
E	Lua Specification Layer	C	Implementation of the embedded Lua scripting interpreter to configure problem data and boundary conditions dynamically.
F	FEM Operator Assembly	D, E	Discretisation and assembly of steady and time-dependent residual and Jacobian terms using `PetscFE` and `PetscDS`.
G	Linear & Transient Solver Integration	F	Integration of Krylov subspace methods, multigrid preconditioning (`gamg`), and `TS` time-marching schemes (Euler, Crank-Nicolson).
H	Code Verification via MMS & Shell	G	Code verification using the Method of Manufactured Solutions and the exact analytical solution of the spherical shell.
I	Numerical Validation & Cross-Validation	H	Numerical validation against reference solutions computed with FreeFEM on re-entrant corner domains (Fichera).
J	Mesh Refinement & Parallel Scaling Study	I	Performance evaluation including uniform/adaptive mesh refinement (`Mmg/ParMmg`) and strong parallel scaling analysis.
K	Post-processing & Visualization	G	Exporting results to `.vtu` and HDF5 formats and generating 3D renders and animations using ParaView.
L	Economic Budget & Impact Analysis	J	Estimation of computational and personal costs, including environmental/social considerations.
M	Final Report Writing	—	Documentation of the full thesis report, mathematical formulation, code listings, and results.
N	Oral Defense Preparation	M	Preparation of presentation slides and materials for the final thesis defense.

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